

PLANT THERAPEUTIC PEPTIDE LAB

PTPLab

User Guide & Platform Documentation

An AI-Driven Platform for Plant-Derived Therapeutic Peptide Discovery, Structural Modelling, Molecular Docking, Molecular Dynamics, and Binding Free-Energy Analysis

Document Version 1.0

Platform Stack: PLM · ESMFold · HADDOCK3 · GROMACS · MM-PBSA

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1. About the Platform

Plant Therapeutic Peptide Lab (PTPLab) is an integrated computational discovery platform designed to accelerate the identification and validation of plant-derived bioactive peptides with therapeutic potential. The platform unifies five distinct computational disciplines - sequence-based bioactivity prediction, protein structure modelling, molecular docking, molecular dynamics simulation, and binding free-energy estimation - into a single, sequential, and reproducible workflow.

The platform is intended for researchers, students, and drug-discovery practitioners who wish to screen candidate peptides computationally before committing to laboratory synthesis and wet-lab validation. Each module in the pipeline produces file outputs that are directly consumable by the following module, while also remaining independently usable through dedicated upload entry points, allowing a user to join the workflow at any stage if they already possess intermediate results (for example, an existing docked complex or a completed molecular dynamics trajectory).

1.1 Scientific Rationale

Therapeutic peptides - short chains of amino acids typically between 3 and 50 residues in length - represent a rapidly growing class of pharmaceutical agents due to their high target specificity, low toxicity, and ease of chemical modification relative to small-molecule drugs. Plants are a particularly rich, under-explored source of such peptides, including defensins, cyclotides, and hydrolysate-derived fragments with documented antibacterial, anticancer, antifungal, antiviral, antiparasitic, anti-inflammatory, and anti-HIV activity.

Identifying which of these peptides are likely to be bioactive, structurally stable, and capable of binding a therapeutic target is traditionally a slow, resource-intensive process. PTPLab addresses this by combining a trained protein language model (PLM) for rapid bioactivity screening with physics-based structural simulation, so that only the most promising candidates are prioritised for experimental follow-up.

1.2 The Five-Stage Pipeline

The platform is organised as a linear pipeline of five modules, summarised below and described in full detail in Sections 2 through 6 of this guide.

Pipeline Stage	Function
Stage 1 - Prediction	Screens a set of candidate peptide sequences for a chosen bioactivity class using a trained protein language model, and ranks them by confidence score and drug-likeness.
Stage 2 - Structure & Receptor	Predicts the three-dimensional structure of the selected peptide using ESMFold, and prepares a cleaned, docking-ready receptor structure from the RCSB Protein Data Bank or a user-supplied file.
Stage 3 - Molecular Docking	Computes the most probable bound orientation (pose) of the peptide against the receptor using HADDOCK3, producing a ranked ensemble of candidate complexes.
Stage 4 - MD Simulation	Subjects the top-ranked docked complex to an all-atom molecular dynamics

Pipeline Stage	Function
	simulation in explicit solvent using GROMACS, assessing the physical stability of the interaction over time.
Stage 5 - MM-PBSA Analysis	Computes the binding free energy (ΔG) of the simulated complex using the MM-PBSA method, providing a quantitative estimate of binding affinity.

DESIGN PRINCIPLE: Every module accepts a manual upload as an alternative to running the preceding pipeline step, so the platform can be entered at any stage without repeating earlier work.

1.3 Underlying Scientific Software

PTPLab does not implement its own physics or machine-learning engines; rather, it orchestrates four established, peer-reviewed scientific tools, each responsible for one stage of the pipeline:

- **Protein Language Model (PLM)** - a sequence-only machine learning model trained to classify short peptide sequences into bioactivity categories without requiring structural information.
- **ESMFold** - a deep learning structure-prediction model (Meta AI) that infers a peptide's three-dimensional fold directly from its amino acid sequence, without the need for homologous templates.
- **HADDOCK3** - a widely cited, data-driven molecular docking suite developed at Utrecht University, used to predict how two molecules - here, a peptide and a receptor protein - associate in three-dimensional space.
- **GROMACS** - a high-performance molecular dynamics engine used to simulate the physical motion of atoms in a system over time, governed by a molecular-mechanics force field.
- **gmx_MMPBSA** - a post-processing tool that applies the Molecular Mechanics Poisson–Boltzmann Surface Area (MM-PBSA) method to a molecular dynamics trajectory to estimate binding free energy.

1.4 Intended Audience

This guide assumes a working familiarity with basic molecular biology and computational chemistry terminology (e.g. amino acids, protein structure, binding affinity). Detailed explanations of every setting, output file, and scientific method are nonetheless provided throughout, so that the platform remains usable by advanced undergraduate and postgraduate researchers without prior expertise in structural bioinformatics.

MODULE 1 Peptide Bioactivity Prediction

This module screens one or more candidate peptide sequences against a trained protein language model to estimate the probability that each sequence exhibits a chosen type of biological activity. It is the entry point of the pipeline and requires only a plain-text sequence file.

2.1 Required Input File

The module accepts a FASTA-format file, recognised by the extensions .fasta, .fa, or .txt. FASTA is the universal plain-text standard for representing biological sequences: each entry begins with a header line starting with the greater-than symbol (>), followed immediately by one or more lines containing the amino acid sequence itself, expressed using the standard single-letter amino acid code (e.g. A for alanine, G for glycine, W for tryptophan).

VALIDATION RULE: A file that does not begin with the '>' character is rejected as invalid, since this is the defining feature of the FASTA format.

2.2 Settings and Their Meaning

Term / Setting	Explanation
Bioactivity Task	Selects which biological activity class the model should screen for. Each candidate sequence is scored against this single class only.
Confidence Threshold	A probability cut-off, ranging from 0.0 to 1.0, below which a peptide is excluded from the results table. A higher threshold yields fewer but more confidently classified candidates; a lower threshold yields a broader, less certain candidate pool.
Min / Max Length	Restricts candidates to a specified amino acid chain-length range. Peptides that are too short may lack a stable secondary structure; peptides that are too long begin to resemble small proteins and lose the pharmacokinetic advantages of a peptide therapeutic.
Net Charge Range	The sum of all positively and negatively charged residues in the sequence at physiological pH. Charge strongly influences membrane interaction, solubility, and - for antimicrobial peptides in particular - the ability to disrupt negatively charged bacterial membranes.
pI Range (Isoelectric Point)	The pH at which the peptide carries no net electrical charge. It is a key determinant of solubility and stability in formulation.
Aromatic Fraction	The proportion of residues that are aromatic (phenylalanine, tryptophan, tyrosine). Aromatic residues contribute to membrane insertion and hydrophobic binding interactions but reduce aqueous solubility when present in excess.
Hydrophobicity	The average hydrophathy of the sequence, calculated from a standard hydrophobicity scale. Extreme hydrophobicity increases membrane permeability but tends to reduce solubility and increase toxicity risk.
Strict Mode	When enabled, any peptide whose properties fall outside the configured filter ranges is marked as failed. When disabled, out-of-range values are shown only as an advisory warning, and the peptide may still be carried forward.

2.3 Bioactivity Classes

The platform screens against seven established therapeutic peptide categories, each abbreviated for use throughout the interface:

Code	Full Name	Therapeutic Relevance
ABP	Antibacterial Peptide	Disrupts or inhibits bacterial cell membranes and metabolic processes; relevant to antibiotic-resistant infections.
ACP	Anticancer Peptide	Selectively induces apoptosis or membrane disruption in malignant cells.
AFP	Antifungal Peptide	Inhibits fungal cell wall or membrane integrity.
AHP	Anti-HIV Peptide	Interferes with viral entry, fusion, or replication machinery specific to HIV.
AIP	Anti-inflammatory Peptide	Modulates immune signalling pathways to reduce inflammatory response.
APP	Antiparasitic Peptide	Active against protozoan or helminthic parasites.
AVP	Antiviral Peptide	Broad-spectrum inhibition of viral entry or replication.

2.4 Generated Output

Running a prediction produces a downloadable comma-separated values file, predictions.csv, containing one row per candidate peptide and columns for the sequence, its bioactivity confidence score, and a calculated drug-likeness percentage. This file can be reopened in any spreadsheet application or reloaded into the platform at a later date.

2.5 Drug-Likeness Evaluation

Every peptide that passes the confidence threshold is additionally scored against seven physicochemical drug-likeness rules, loosely analogous to Lipinski's Rule of Five used for small-molecule drugs, but adapted for peptide therapeutics:

Term / Setting	Explanation
Molecular Weight	Total mass of the peptide in Daltons (Da), summed from constituent residues. A ceiling of 2000 Da is applied, beyond which oral and systemic bioavailability typically decline sharply.
Net Charge	Recommended range of -5 to +5. Extreme charge states are associated with poor membrane permeability and increased off-target electrostatic binding.
Hydrogen-Bond Donors (HBD)	Count of donor groups (serine, threonine, asparagine, glutamine residues). A ceiling of 10 mirrors established permeability guidelines.

Term / Setting	Explanation
Hydrogen-Bond Acceptors (HBA)	Count of acceptor groups (aspartate, glutamate, serine, threonine residues), capped at 20.
Aromatic Fraction	Recommended ceiling of 0.35 (35% of residues), beyond which aggregation and poor solubility become more likely.
Hydrophobicity (GRAVY-style score)	Recommended range of -2 to +3, balancing membrane permeability against aqueous solubility.
Length	Recommended range of 3–50 amino acids, the conventional definition of a peptide as opposed to a full protein.

The percentage of these seven rules that a candidate satisfies is reported as its overall drug-likeness score. Scores of 80% or above are shaded green (favourable), 60–79% are shaded amber (borderline), and below 60% are shaded red (unfavourable).

2.6 Visual Analytics

- **Confidence Score Chart** - a ranked bar chart of the top-N peptides by bioactivity confidence, used to quickly identify the strongest candidates.
- **Drug-Likeness Distribution** - a histogram of drug-likeness scores across the full candidate set, with a reference line at the 70% threshold.
- **Chemical Space Map (3D PCA)** - a three-dimensional Principal Component Analysis projection of all candidates based on seven physicochemical descriptors (length, molecular weight, charge, hydrophobicity, aromaticity, proline content, glycine content). Peptides that cluster closely together share similar physicochemical character; colour encodes bioactivity score and marker size encodes sequence length. This visualisation requires a minimum of three candidate peptides to be meaningful.
- **Physicochemical Radar Profile** - generated for a single selected peptide during detailed evaluation, comparing its length, charge, pI, aromaticity, and hydrophobicity against the user-configured filter ranges on a normalised radial scale.

MODULE 2 Structure Prediction & Receptor Preparation

This module establishes the three-dimensional atomic coordinates required for docking: one structure for the candidate peptide (the ligand) and one for the biological target (the receptor).

3.1 Peptide Structure - ESMFold

ESMFold is a deep learning model that predicts a protein or peptide's three-dimensional backbone and side-chain conformation directly from its amino acid sequence. Unlike classical homology modelling, ESMFold does not require a structurally similar template to already exist in a database; it instead relies on evolutionary and structural patterns learned from hundreds of millions of protein sequences during training. This makes it particularly well suited to short, non-standard peptides - including plant-derived peptides - for which no crystallographic structure is likely to exist.

The user may alternatively bypass ESMFold entirely by uploading a pre-existing peptide structure file in PDB format, for example one obtained from a prior simulation, a published crystal structure, or an NMR ensemble.

3.2 Receptor Preparation

The receptor is the macromolecular target - typically a disease-relevant protein - that the candidate peptide is hypothesised to bind. Receptor structures may be obtained in either of two ways:

- **Download from RCSB PDB** - by supplying a four-character PDB identifier (for example 1YCR for MDM2, or 1G5M for BCL-2), the platform retrieves the corresponding experimentally-determined structure directly from the RCSB Protein Data Bank, the primary global repository for macromolecular structural data.
- **Direct Upload** - a locally held receptor PDB file may be uploaded in place of a database download.

3.3 Automated Receptor Cleaning

Raw structures deposited in the Protein Data Bank frequently contain crystallographic artefacts that are not appropriate for docking or simulation, including bound ligands, co-crystallised water molecules, alternate conformations, and - in the case of protein–DNA complexes - nucleic acid chains. The platform automatically removes these elements upon download or upload, retaining only standard protein residues, and reports the number of atoms retained together with any non-standard residues that were removed for transparency.

3.4 Chain Selection

Where a receptor structure contains more than one polypeptide chain (for example, a homodimer or a multi-subunit complex), the platform allows the user to explicitly select which chain, or combination of chains, should be retained for docking. This is important because including an unintended chain can introduce steric clashes or bias the docking search toward an irrelevant interface.

3.5 Output Files

File	Format	Description
peptide.pdb	PDB	Three-dimensional coordinates of the folded or uploaded peptide.
receptor.pdb / receptor_clean.pdb	PDB	Cleaned receptor structure, stripped of ligands, water, and non-standard residues, ready for docking.

MODULE 3 Molecular Docking (HADDOCK3)

Molecular docking computationally predicts the preferred spatial orientation of one molecule (the peptide) when bound to another (the receptor), and estimates the strength and geometry of that interaction. This module uses HADDOCK3 (High Ambiguity Driven protein-protein DOCKing, version 3), a data-driven docking engine that is distinguished from purely geometric or 'blind' docking methods by its ability to incorporate biochemical and biophysical information to steer the search toward biologically realistic interfaces.

4.1 Docking Settings and Their Meaning

Term / Setting	Explanation
Sampling Poses	The number of independent, randomly-oriented rigid-body starting configurations generated at the beginning of the docking run. A higher sampling value more thoroughly explores the space of possible binding orientations at the cost of increased computation time.
CPU Cores	The number of processor cores allocated to parallelise the docking calculation. Increasing this value reduces wall-clock run time on multi-core hardware.
Top Models for Refinement	The number of best-scoring poses, selected after initial rigid-body docking and clustering, that are carried forward into the computationally expensive flexible-refinement stage. Restricting this number keeps runtime manageable while still refining only the most promising candidates.

4.2 The HADDOCK3 Workflow

Each docking run proceeds through a fixed sequence of stages, visible in the platform as a live progress tracker:

- **Topology (topoaa)** - generates the molecular topology and force-field parameters required for all subsequent stages.
- **Rigid Body Docking** - performs the initial randomised docking search, treating both molecules as rigid bodies, to generate the full sampling pool of candidate poses.
- **CAPRI Evaluation** - scores each generated pose using CAPRI (Critical Assessment of PRedicted Interactions) quality metrics, a community-standard framework for evaluating docking accuracy.
- **Clustering (FCC)** - groups structurally similar poses together using Fraction of Common Contacts clustering, reducing redundancy in the candidate pool.
- **Selection of Top Clusters** - retains only the best-ranked clusters for further refinement, as configured by the Top Models setting.
- **Flexible Refinement** - relaxes side-chain and limited backbone flexibility at the binding interface to correct steric clashes introduced by rigid-body docking.
- **Energy Minimisation Refinement (emref)** - performs a final short energy minimisation to relax the refined complex into a locally optimal geometry.
- **Final CAPRI Evaluation** - re-scores the refined poses to produce the final ranked output.

4.3 Interpreting the HADDOCK Score

The HADDOCK score is a weighted combination of van der Waals energy, electrostatic energy, desolvation energy, and restraint violation terms. It is a relative, unitless scoring function: lower (more negative) values indicate a more energetically favourable predicted interaction.

HADDOCK Score	Interpretation
Below -50	Strong predicted binding interaction.
-30 to -50	Moderate predicted binding interaction.
Above -30	Weak or uncertain predicted binding interaction; further validation is recommended.

4.4 Alternative Entry Points

In addition to running the full HADDOCK3 pipeline, this module provides two alternative pathways: direct upload of an already-docked complex structure (bypassing docking entirely), and loading of results from a previously completed, external HADDOCK3 run directory.

4.5 Output Files

File	Format	Description
docking.toml	TOML (config)	The configuration file defining the docking protocol and parameters submitted to HADDOCK3.
capri_ss.tsv	Tab-separated values	Per-pose scoring table including HADDOCK score, electrostatics, van der Waals energy, buried surface area (BSA), and DOCKQ quality metric.
docked_complex.pdb	PDB	The top-ranked docked peptide–receptor complex, used as the direct input to the MD Simulation module.

MODULE 4 Molecular Dynamics Simulation (GROMACS)

A docking result represents a single, static snapshot of a possible binding pose. Molecular dynamics (MD) simulation tests whether that pose is physically stable by numerically integrating the equations of motion for every atom in the system over a period of simulated time, under an explicit solvent environment and a defined temperature and pressure. This module performs a complete MD workflow using GROMACS, one of the most widely used and validated molecular dynamics engines in computational biophysics.

5.1 Simulation Settings and Their Meaning

Term / Setting	Explanation
Force Field	The mathematical model and parameter set that defines how atoms interact - bond lengths, angles, torsions, van der Waals radii, and partial charges. AMBER99SB-ILDN is the platform default and is well validated for protein and peptide systems; AMBER03, CHARMM27, and OPLS-AA are provided as alternatives, each with slightly different parameterisation philosophies suited to particular molecule classes.
Water Model	The explicit representation used for solvent water molecules. TIP3P (the default) is a computationally efficient three-site model widely paired with AMBER force fields; TIP4P and SPC/E offer improved accuracy at moderately higher computational cost.
MD Length	The total simulated (not real) time duration of the production run, expressed in nanoseconds (ns). Longer simulations more thoroughly sample the conformational behaviour of the complex but require proportionally more computation time.
Time Step	The discrete increment of simulated time, in picoseconds (ps), advanced at each integration step. A time step of 0.002 ps (2 femtoseconds) is standard when bonds involving hydrogen are constrained, as is done here; smaller time steps increase numerical stability at the cost of requiring more steps to reach the same total simulated time.
CPU Cores	The number of processor threads allocated to the GROMACS integration engine, directly affecting wall-clock run time.

5.2 The Simulation Workflow

The underlying script executes a standard, published MD protocol in the following sequence:

- **Topology Generation (pdb2gmx)** - converts the docked complex's atomic coordinates into a GROMACS-compatible topology, applying the selected force field.
- **Box Construction & Solvation** - places the complex inside a dodecahedral simulation box with a minimum 1.0 nanometre buffer to the box edge, then fills the remaining volume with explicit water molecules of the selected model.
- **Ion Addition** - adds sodium and chloride counter-ions to neutralise the net charge of the system, replicating physiological ionic strength.
- **Energy Minimisation** - removes steric clashes and unfavourable geometries introduced during system construction by iteratively adjusting atomic positions toward a local energy minimum, using the steepest-descent algorithm.

- **NVT Equilibration** - equilibrates the system temperature at 300 K under constant volume, with position restraints on the solute to allow the solvent to relax first.
- **NPT Equilibration** - equilibrates system pressure at 1 bar under constant temperature, allowing the box volume and solvent density to reach realistic physiological values.
- **Production MD** - the unrestrained simulation run whose duration is set by the MD Length setting; this is the phase from which all subsequent analysis is derived.
- **Trajectory Analysis** - computes RMSD, RMSF, and radius of gyration from the production trajectory (see Section 5.4).
- **MM-PBSA Calculation** - automatically invoked at the end of the MD run; described fully in Module 5.

5.3 Software Requirements

PREREQUISITE: This module requires both GROMACS (gmx) and gmx_MMPBSA to be installed and accessible within a conda environment named mmpbsa_env on the host machine. Installation commands are provided in Section 8 of this guide.

5.4 Interpreting Trajectory Analysis

Term / Setting	Explanation
RMSD (Root-Mean-Square Deviation)	Measures how much the complex's backbone structure deviates over time from its initial conformation. A mean RMSD below approximately 0.2 nanometres indicates a structurally stable complex; sustained higher values suggest conformational drift or instability of the binding pose.
RMSF (Root-Mean-Square Fluctuation)	Measures the average positional fluctuation of each individual residue across the trajectory. High RMSF values identify flexible loop regions; low values identify rigid, structurally constrained regions such as the binding interface.
Radius of Gyration (Rg)	Measures the overall compactness of the complex. A stable Rg over time indicates the complex is not unfolding or dissociating during the simulation.

5.5 Output Files

File	Format	Description
topol.top	GROMACS topology	Complete molecular topology of the solvated, ionised system.
em.gro, nvt.gro, npt.gro, md.gro	GROMACS coordinate	Atomic coordinates at the end of each simulation stage.
md.xtc / md_fit.xtc	Compressed trajectory	The full time-resolved atomic trajectory from the production run.
rmsd.xvg, rmsf.xvg, gyrate.xvg	XVG (plain text)	Time-series and per-residue data underlying the stability plots described above.
FINAL_MMPBSA.dat / .csv	Text / CSV	Binding free-energy results, consumed directly by Module 5.

MODULE 5 MM-PBSA Binding Free-Energy Analysis

The final module quantifies the strength of the peptide–receptor interaction by estimating its binding free energy (ΔG) using the Molecular Mechanics Poisson–Boltzmann Surface Area (MM-PBSA) method, calculated across the production molecular dynamics trajectory via `gmx_MMPBSA`. Unlike the HADDOCK score, which is a relative, unitless docking score, MM-PBSA produces an approximate physical energy value expressed in kilocalories per mole (kcal/mol), allowing more direct comparison between different candidate peptides.

6.1 Energy Decomposition Terms

Term / Setting	Explanation
ΔG Binding (DELTA TOTAL)	The total estimated binding free energy: the net energetic favourability of the peptide binding to the receptor relative to both being unbound in solvent. More negative values indicate stronger predicted binding.
Van der Waals Energy	The contribution from short-range dispersion and steric packing interactions at the binding interface.
Electrostatic Energy	The contribution from Coulombic interactions between charged and polar atoms across the interface.
Polar Solvation Energy	The energetic penalty of desolvating polar groups upon binding, computed via the Poisson–Boltzmann equation.
Non-Polar Solvation Energy	The energetic contribution associated with burial of non-polar (hydrophobic) surface area upon complex formation.

6.2 Interpreting the Binding Free Energy

Binding Free Energy	Interpretation
$\Delta G \leq -10$ kcal/mol	Very strong binder - affinity in the nanomolar range.
$\Delta G -7$ to -10 kcal/mol	Strong binder.
$\Delta G -5$ to -7 kcal/mol	Moderate binder - affinity in the micromolar range.
$\Delta G > -5$ kcal/mol	Weak binder; the peptide is unlikely to form a stable complex under physiological conditions.

SCIENTIFIC CAVEAT: MM-PBSA values are approximate estimates derived from a finite simulation window and an implicit solvent model; they should be interpreted as a ranking tool for comparing candidates rather than as an exact experimental binding constant.

6.3 Alternative Entry Point

Users who have already generated MD output files using an independent, external GROMACS and `gmx_MMPBSA` workflow may upload the resulting `FINAL_MMPBSA.dat`, `FINAL_MMPBSA.csv`, and `.xvg` trajectory files directly, allowing this module's visual analytics to be used without repeating the simulation within the platform.

6.4 Output Files Available for Download

File	Format	Description
docked_complex.pdb	PDB	The final docked structure carried through the pipeline.
FINAL_MMPBSA.dat	Plain text	Full, human-readable MM-PBSA energy report.
FINAL_MMPBSA.csv	CSV	Per-frame MM-PBSA energy values, suitable for external statistical analysis.
rmsd.xvg, rmsf.xvg, gyrate.xvg	XVG (plain text)	Underlying trajectory analysis data.
md.gro, topol.top	GROMACS coordinate / topology	Final simulation structure and system topology, for continuation in an external GROMACS session.
capri_results.tsv	Tab-separated values	Docking-stage scoring table, retained for full pipeline traceability.
predictions.csv	CSV	Original bioactivity prediction table from Module 1.

7. Complete File Format Reference

The table below consolidates every file format produced or consumed across the full pipeline, for quick reference.

Extension	Format Name	Role in the Platform
.fasta / .fa / .txt	FASTA sequence	Input format for candidate peptide sequences at Module 1.
.pdb	Protein Data Bank	Standard format for atomic 3-D coordinates; used for peptides, receptors, and docked complexes throughout Modules 2–4.
.csv	Comma-separated values	Tabular prediction, drug-likeness, and MM-PBSA results, openable in any spreadsheet application.
.tsv	Tab-separated values	Docking scoring tables (CAPRI evaluation output).
.toml	Tom's Obvious Minimal Language	Human-readable configuration file format used to define the HADDOCK3 docking protocol.
.top / .itp	GROMACS topology	Defines atom types, bonds, angles, and force-field parameters for the simulated system.
.gro	GROMACS coordinate	Atomic coordinates and box dimensions at a given simulation stage.
.tpr	GROMACS binary run input	Pre-processed, portable binary file combining topology, coordinates, and simulation parameters, used to launch a GROMACS run.
.xtc	Compressed trajectory	Space-efficient, time-resolved record of atomic positions across a simulation.
.mdp	GROMACS parameter file	Plain-text file specifying simulation parameters (integrator, time step, thermostat, barostat) for each MD stage.
.xvg	XmGrace plot data	Plain-text, two-column time-series data, used here for RMSD, RMSF, and radius of gyration plots.
.dat	MM-PBSA report	Full, formatted binding free-energy decomposition report generated by gmx_MMPBSA.
.ndx	GROMACS index	Defines named atom groups (e.g. Receptor, Ligand) used to direct

Extension	Format Name	Role in the Platform
		GROMACS analysis commands.

8. System Requirements & Installation

The platform depends on two isolated conda environments, separating the docking toolchain from the molecular dynamics toolchain to avoid dependency conflicts.

8.1 Docking Environment (haddock_env)

```
conda create -n haddock_env python=3.10 conda activate haddock_env conda install -c conda-forge haddock3 streamlit  
pip install fair-esm py3dmol plotly scikit-learn biopython
```

8.2 Molecular Dynamics Environment (mmpbsa_env)

```
conda create -n mmpbsa_env python=3.10 conda activate mmpbsa_env conda install -c conda-forge gromacs pip install  
gmx_MMPBSA ambertools
```

8.3 Launching the Platform

```
streamlit run app.py
```

9. Glossary of Technical Terms

Term / Setting	Explanation
Bioactivity	The capacity of a molecule to produce a measurable biological or pharmacological effect.
BSA (Buried Surface Area)	The molecular surface area concealed from solvent at the interface of a bound complex; larger values generally indicate a more extensive binding interface.
CAPRI	Critical Assessment of PRedicted Interactions - a community benchmark framework and associated quality metrics for evaluating docking predictions.
ΔG (Delta G)	Gibbs free energy change; in this context, the free-energy difference between the bound and unbound states of a peptide–receptor pair.
DOCKQ	A single continuous quality score (0 to 1) summarising the accuracy of a predicted docking pose relative to a reference structure.
Force Field	A parameterised mathematical model describing the potential energy of a molecular system as a function of atomic positions.
Isoelectric Point (pI)	The pH value at which a molecule carries no net electrical charge.
PLM (Protein Language Model)	A machine learning model trained on large corpora of protein sequences, capable of predicting biological properties directly from sequence.
Pose	A single candidate three-dimensional binding orientation generated during docking.
RMSD	Root-Mean-Square Deviation; a measure of structural difference between two conformations of the same molecule.
RMSF	Root-Mean-Square Fluctuation; a per-residue measure of positional variability across a simulation trajectory.
Radius of Gyration (Rg)	A measure of the overall compactness of a molecular structure.
Topology	In molecular simulation, the complete definition of a molecule's atoms, bonds, and associated force-field parameters.

10. References & Attribution

PTPLab is built entirely upon open-source and academically published scientific software. Users publishing results obtained using this platform are strongly encouraged to cite the original authors of each underlying tool.

Tool	Reference
HADDOCK3	https://github.com/haddocking/haddock3
ESMFold	https://esmatlas.com
GROMACS	https://www.gromacs.org
gmx_MMPBSA	https://github.com/Valdes-Tresanco-MS/gmx_MMPBSA

Plant Therapeutic Peptide Lab - v1.0

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